

# Notes: Abreu and Brunnermeier (Econometrica, 2003)

## Bubbles and Crashes

### Contents

|           |                                                                    |          |
|-----------|--------------------------------------------------------------------|----------|
| <b>1</b>  | <b>Big picture</b>                                                 | <b>2</b> |
| <b>2</b>  | <b>Incremental contribution of the paper</b>                       | <b>2</b> |
| <b>3</b>  | <b>Model objects and measurement</b>                               | <b>3</b> |
| 3.1       | Price path and fundamental value . . . . .                         | 3        |
| 3.2       | Sequential awareness . . . . .                                     | 3        |
| 3.3       | Critical mass and trading constraints . . . . .                    | 3        |
| 3.4       | Hazard rates and sellout condition . . . . .                       | 3        |
| <b>4</b>  | <b>Models and theoretical specifications</b>                       | <b>4</b> |
| 4.1       | Bubble price process . . . . .                                     | 4        |
| 4.2       | Posterior over the bubble starting date . . . . .                  | 4        |
| 4.3       | Exogenous crash hazard . . . . .                                   | 4        |
| 4.4       | Endogenous crash condition . . . . .                               | 4        |
| 4.5       | Equilibrium logic . . . . .                                        | 5        |
| <b>5</b>  | <b>Identification and theoretical design logic</b>                 | <b>5</b> |
| 5.1       | What plays the role of identification . . . . .                    | 5        |
| 5.2       | Why sequential awareness matters . . . . .                         | 5        |
| 5.3       | Why this is not irrationality . . . . .                            | 5        |
| 5.4       | Main limitations . . . . .                                         | 6        |
| <b>6</b>  | <b>Section-by-section study guide</b>                              | <b>6</b> |
| <b>7</b>  | <b>Figures and theoretical result blocks</b>                       | <b>6</b> |
| 7.1       | Figures 1 and 2: price paths and sequential awareness . . . . .    | 6        |
| 7.2       | Figures 3 and 4: exogenous crash versus endogenous crash . . . . . | 7        |
| 7.3       | Theoretical result block 1: synchronization risk . . . . .         | 8        |
| 7.4       | Theoretical result block 2: riding the bubble . . . . .            | 8        |
| 7.5       | Theoretical result block 3: endogenous crash . . . . .             | 8        |
| 7.6       | Theoretical result block 4: comparative statics . . . . .          | 9        |
| <b>8</b>  | <b>Detailed result map</b>                                         | <b>9</b> |
| <b>9</b>  | <b>Short summary</b>                                               | <b>9</b> |
| <b>10</b> | <b>Conclusion</b>                                                  | <b>9</b> |

|                                                   |    |
|---------------------------------------------------|----|
| 10.1 Core conclusion . . . . .                    | 9  |
| 10.2 Economic intuition . . . . .                 | 10 |
| 10.3 Incremental contribution revisited . . . . . | 10 |
| 10.4 Implications for financial markets . . . . . | 10 |
| 10.5 Limitations . . . . .                        | 10 |
| 10.6 Final takeaway . . . . .                     | 10 |

# 1 Big picture

- **Question.** Why can rational arbitrageurs fail to eliminate an asset-price bubble even when they know the asset is overpriced?
- **Core answer.** Bubbles can persist because informed arbitrageurs become aware of mispricing at different times and do not know exactly how many other arbitrageurs are already aware. This creates a synchronization problem.
- **Key mechanism.** An arbitrageur who attacks the bubble too early may suffer losses while prices continue rising. Since other arbitrageurs may not yet be aware, the best response can be to ride the bubble for a while and exit only later.
- **Central friction.** Arbitrageurs face holding/short-selling limits and cannot coordinate. They also face an uncertain crash date. Even if they all recognize overpricing eventually, they do not all know when enough others recognize it.
- **Main result.** The bubble crashes endogenously only after enough arbitrageurs have become aware and after a predictable delay. Rational arbitrageurs ride the bubble rather than immediately arbitraging it away.
- **Why this matters.** The paper explains why bubbles can grow and then crash abruptly without relying on irrational investors alone. Irrational or behavioral demand may start the bubble, but rational arbitrageurs can rationally allow it to continue.
- **Broader implication.** Market efficiency can fail when correcting mispricing requires coordinated timing. Private information about mispricing is not enough; arbitrage must also be synchronized.

# 2 Incremental contribution of the paper

- **Rational bubble survival.** Earlier limits-to-arbitrage work emphasized capital constraints, noise-trader risk, or fundamental risk. Abreu and Brunnermeier add *synchronization risk*: even informed arbitrageurs may delay because they cannot coordinate the timing of their trades.
- **Dynamic asymmetric-awareness model.** The paper introduces a dynamic model in which arbitrageurs become aware of the bubble sequentially. This is different from standard asymmetric-information models in which all agents know the information structure from the beginning.
- **Coordination without communication.** The paper shows that the problem is not only whether arbitrageurs know the price is wrong; it is whether they know that enough other arbitrageurs know. This anticipates later common-knowledge and global-games views of financial crises.
- **Endogenous crash timing.** The model generates a crash time endogenously from arbitrageur selling strategies, not simply from an exogenous fundamental shock.
- **Riding-the-bubble logic.** The paper formalizes why a rational arbitrageur may optimally hold, or even buy, a bubble asset despite knowing it is overvalued: expected continuation gains can exceed crash risk until the endogenous sellout date approaches.
- **Unification of bubbles and crashes.** The paper explains both the persistence phase and the crash phase in one model. The same synchronization mechanism that lets the bubble survive creates the abrupt crash once enough informed traders sell.
- **Resolution of symmetric-information discontinuity.** Under symmetric information, arbitrage removes bubbles. Under small asymmetric awareness, bubbles can persist, but the equilibrium converges smoothly to the no-bubble outcome as the degree of asymmetry disappears.
- **Asset-pricing and macro-finance relevance.** The framework gives a rational foundation for delayed correction in technology booms, credit booms, real estate bubbles, and asset-market manias where sophisticated investors privately recognize overvaluation but cannot coordinate exits.

### 3 Model objects and measurement

#### 3.1 Price path and fundamental value

- The observed bubble price follows an upward path, written in the paper as  $p_t = e^{gt}$ .
- After the crash, the asset trades at a lower fundamental value, represented in Figure 1 as a downward-adjusted path such as  $(1 - \beta(t - t_0))p_t$ .
- $t_0$  is the random time when the bubble begins from the perspective of informed arbitrage.
- $\bar{\tau}$  is the maximum life span of the bubble, after which it bursts for exogenous reasons.

**Interpretation:** The price path gives arbitrageurs a reason to delay. If the price keeps growing, selling too early can be costly even when the asset is overvalued.

#### 3.2 Sequential awareness

- Arbitrageurs do not all become aware of the bubble at the same time.
- Each arbitrageur  $t_i$  learns about the mispricing at a private awareness date.
- Awareness diffuses gradually over an interval of length  $\eta$ .
- An arbitrageur who becomes aware does not know exactly when the first arbitrageurs became aware or how many others are currently aware.

**Interpretation:** Sequential awareness creates the central coordination problem. The agent knows there is a bubble but not whether the market has reached the critical mass of informed arbitrageurs needed to attack it.

#### 3.3 Critical mass and trading constraints

- A critical mass of arbitrageurs must sell to burst the bubble endogenously.
- The parameter  $\kappa$  represents the fraction of arbitrageurs that must be aware before enough selling pressure can be generated.
- Arbitrageurs face limits on long and short positions; without such constraints, a single arbitrageur could take an unbounded position and collapse the bubble.
- Each arbitrageur can hold, buy, sell, or stay out of the market, but she must decide without observing the full awareness distribution.

**Interpretation:** Limited arbitrage and gradual awareness jointly create the delayed-crash equilibrium. The model is not simply about short-sale constraints; it is about strategic timing under incomplete higher-order beliefs.

#### 3.4 Hazard rates and sellout condition

- An informed arbitrageur compares the benefit of remaining in the bubble to the crash risk.
- The benefit is the expected excess growth rate of the bubble price over the safe return.
- The cost is the hazard-adjusted expected capital loss from a crash.
- Sellout occurs when the hazard cost reaches the continuation benefit.

**Interpretation:** The figures translate the equilibrium into hazard-rate diagrams. Exogenous and endogenous crashes are both driven by a comparison between hazard risk and the benefit of riding the bubble.

## 4 Models and theoretical specifications

### 4.1 Bubble price process

The bubble price rises according to

$$p_t = e^{gt}.$$

A post-crash fundamental value is lower than the bubble price. In the paper's illustrative path, the post-crash value falls relative to the bubble price at a rate governed by  $\beta$  and the time since the bubble began.

**Interpretation:** The gap between  $p_t$  and the post-crash value creates potential arbitrage profit, but the rising bubble price creates a cost to selling too soon.

### 4.2 Posterior over the bubble starting date

When arbitrageur  $t_i$  becomes aware, she forms a posterior over  $t_0$  on the interval  $[t_i - \eta, t_i]$ . The paper uses a truncated distribution of the form

$$\Phi(t_0|t_i) = \frac{e^{\lambda\eta} - e^{\lambda(t_i - t_0)}}{e^{\lambda\eta} - 1}.$$

**Interpretation:** An arbitrageur's inference problem is relative: she knows her own awareness date, but not how early or late she is in the diffusion of awareness.

### 4.3 Exogenous crash hazard

If the bubble bursts at the exogenous deadline  $t_0 + \bar{\tau}$ , then an arbitrageur's hazard rate at delay  $\tau$  has the form

$$h(t_i + \tau|t_i) = \frac{\lambda}{1 - e^{-\lambda(\bar{\tau} - \tau)}}.$$

**Interpretation:** The hazard rate rises as the exogenous deadline approaches. The arbitrageur optimally exits when the hazard cost becomes large enough relative to the continuation benefit.

### 4.4 Endogenous crash condition

In the endogenous-crash case, enough arbitrageurs have become aware and coordinated selling can burst the bubble. The equilibrium sellout delay  $\tau^*$  solves a condition like

$$\frac{\lambda}{1 - e^{-\lambda\eta\kappa}} = \frac{g - r}{\beta(\eta\kappa + \tau^*)}.$$

**Interpretation:** The left-hand side is the effective crash hazard generated by the awareness process. The right-hand side is the benefit-cost threshold from riding the bubble. Their intersection determines the crash time.

## 4.5 Equilibrium logic

- Early informed arbitrageurs hold or ride the bubble because they expect further price increases and do not yet expect enough coordinated selling.
- As time passes, each arbitrageur becomes more confident that enough other arbitrageurs are aware.
- Once the endogenous sellout point is reached, selling becomes self-confirming and the bubble crashes.
- The crash is predictable in equilibrium to informed arbitrageurs only through their posterior beliefs, not through a publicly known announcement.

**Interpretation:** The model creates an equilibrium in which rational traders knowingly tolerate mispricing until the timing problem is solved.

## 5 Identification and theoretical design logic

### 5.1 What plays the role of identification

This is a theoretical paper. The identification logic is conceptual rather than empirical. The model isolates one force - synchronization risk - by assuming rational arbitrageurs, common model structure, and a stylized bubble process.

**Interpretation:** The paper's theoretical design is meant to show that rationality alone does not guarantee immediate arbitrage. The missing ingredient is common timing knowledge.

### 5.2 Why sequential awareness matters

- If all arbitrageurs learned simultaneously, the bubble would be attacked immediately.
- If no arbitrageur faced position limits, one arbitrageur could collapse the bubble alone.
- If the bubble had no expected continuation return, riding would not be attractive.
- If awareness dispersion vanished, the equilibrium converges toward the symmetric-information no-bubble outcome.

**Interpretation:** These counterfactuals clarify the mechanism: the bubble survives only because awareness is dispersed, arbitrage is limited, and coordination is required.

### 5.3 Why this is not irrationality

- Arbitrageurs understand the asset is overpriced.
- They understand the crash risk.
- They understand other arbitrageurs are gradually becoming aware.
- They still rationally choose to wait because early selling is individually costly.

**Interpretation:** The model converts a seemingly irrational behavior - knowingly riding a bubble - into a rational equilibrium outcome.

## 5.4 Main limitations

- The paper is theoretical and does not estimate awareness processes empirically.
- The bubble price path is stylized and exogenous before the crash.
- Position limits are imposed rather than derived from balance-sheet frictions.
- The model abstracts from heterogeneous beliefs beyond the awareness process.
- Communication among arbitrageurs is excluded, although real institutions may partially coordinate through prices, news, or networks.

**Interpretation:** These assumptions are useful because they isolate the synchronization mechanism. The model is a mechanism paper, not a full empirical model of every bubble episode.

## 6 Section-by-section study guide

- **Introduction.** Motivates the puzzle: sophisticated investors sometimes ride bubbles even when they recognize overpricing. The paper argues that delayed arbitrage is rational when awareness is asynchronous.
- **Related literature.** Distinguishes the model from static asymmetric-information bubbles, noise-trader-risk models, and standard limits-to-arbitrage arguments.
- **Model.** Defines the bubble price path, the crash value, awareness diffusion, posterior beliefs, position limits, and arbitrageur strategies.
- **Equilibrium analysis.** Derives when arbitrageurs exit in exogenous and endogenous crash cases and shows how a unique symmetric equilibrium arises.
- **Extensions and discussion.** Explains why synchronization risk can rationalize bubble persistence and sudden collapse and how the mechanism applies to real-world asset booms.
- **Appendix/proofs.** Verifies the sellout conditions, hazard comparisons, uniqueness, and equilibrium properties.

## 7 Figures and theoretical result blocks

### 7.1 Figures 1 and 2: price paths and sequential awareness

**Read:**

- Figure 1 shows the rising bubble price  $p_t = e^{gt}$  and the lower post-crash value path.
- The start of the bubble is random at  $t_0$ ; some arbitrageurs become aware at different times; all arbitrageurs are aware by  $t_0 + \eta$ .
- The bubble can burst endogenously after enough arbitrageurs become aware or exogenously at the maximum life span  $t_0 + \bar{\tau}$ .
- Figure 2 shows that arbitrageurs have different posterior distributions over  $t_0$  depending on their private awareness date.

**Detailed interpretation:** Figure 1 gives the model's economic environment: a bubble price continues to rise even though its post-crash value is lower. This creates the possibility that a rational arbitrageur knows the price is wrong but still earns money by waiting. Figure 2 gives the information structure: arbitrageurs know their own awareness date but cannot tell whether they are early or late relative to other arbitrageurs. The figures together show why mispricing is not sufficient for immediate arbitrage.

**Economic message:** The bubble survives because correcting it requires both information and synchronized timing. Private knowledge of overpricing is not enough.

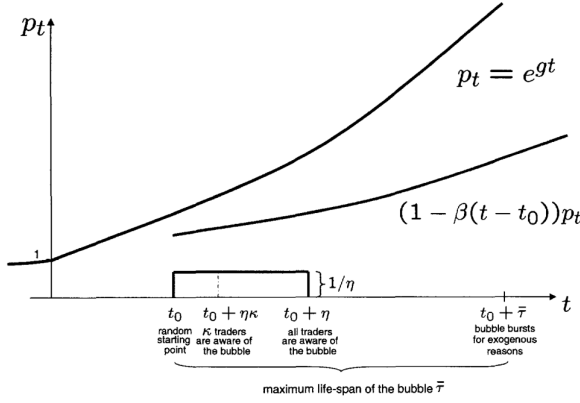


FIGURE 1.—Illustration of price paths.

(a) Figure 1 - Illustration of price paths

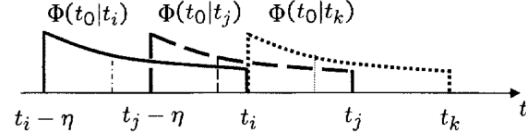


FIGURE 2.—Sequential awareness.

(b) Figure 2 - Sequential awareness

## 7.2 Figures 3 and 4: exogenous crash versus endogenous crash

**Read:**

- Figure 3 shows the exogenous-crash hazard rising as the maximum crash date approaches.
- The horizontal line is the cost-benefit threshold from the sellout condition.
- The intersection determines the exit date when the crash is exogenous.
- Figure 4 shows the endogenous-crash equilibrium where a constant effective crash hazard intersects a declining continuation-benefit schedule.
- The equilibrium sellout date  $\tau^*$  is where waiting no longer dominates selling.

**Detailed interpretation:** Figure 3 describes the case where the bubble will end for reasons outside arbitrageurs' selling. As time passes, the exogenous crash hazard rises, so exiting becomes attractive. Figure 4 is more central to the paper's contribution: the crash is caused by arbitrageurs' collective selling once enough of them have become aware. The intersection condition formalizes why all arbitrageurs wait for a while and then exit in a coordinated way.

**Economic message:** The difference between Figure 3 and Figure 4 is the difference between an outside crash and a crash generated by arbitrage itself. The model's key contribution is the endogenous crash.



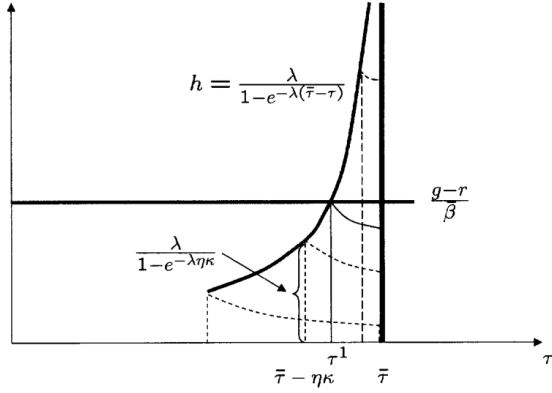


FIGURE 3.—Exogenous crash.

(a) Figure 3 - Exogenous crash

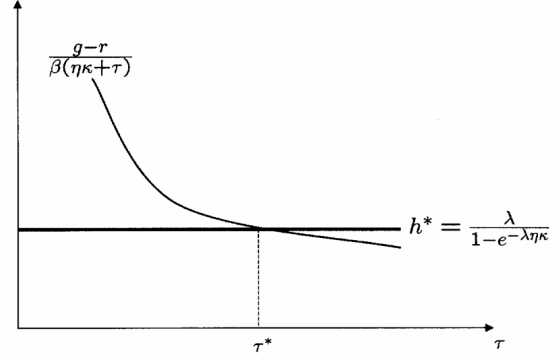


FIGURE 4.—Endogenous crash.

(b) Figure 4 - Endogenous crash

### 7.3 Theoretical result block 1: synchronization risk

- A bubble cannot be eliminated by one small arbitrageur when positions are constrained.
- A coordinated mass of arbitrageurs is needed to burst the bubble.
- Each arbitrageur is uncertain about how many others are already aware.
- The individual best response is to wait until the posterior probability of sufficient awareness becomes high enough.

**Interpretation:** Synchronization risk is different from fundamental risk. The arbitrageur may know the asset is overpriced but not know when others will attack.

### 7.4 Theoretical result block 2: riding the bubble

- The bubble price grows at rate  $g$ .
- The arbitrageur can earn continuation gains by staying in the bubble.
- Early shorting is costly because the price can continue rising before the crash.
- The optimal strategy is therefore not always immediate selling; it can be delayed exit.

**Interpretation:** Riding the bubble is rational when expected price growth exceeds the crash-risk-adjusted cost of waiting.

### 7.5 Theoretical result block 3: endogenous crash

- The crash occurs when enough informed arbitrageurs sell.
- The sellout date is determined by the intersection of a hazard term and a continuation-benefit term.
- The crash is sudden because many arbitrageurs face the same strategic sellout logic.
- The bubble's endpoint is therefore generated inside the arbitrage game.

**Interpretation:** Endogenous crashes are abrupt because the coordination problem resolves discontinuously once the threshold is crossed.

## 7.6 Theoretical result block 4: comparative statics

- Faster awareness diffusion lowers the time during which the bubble can survive.
- A higher bubble growth rate makes riding more attractive and can delay exit.
- Tighter position limits make synchronization more important.
- A lower crash loss or lower post-crash gap weakens the incentive to exit early.
- As awareness dispersion goes to zero, the equilibrium converges toward immediate arbitrage/no bubble.

**Interpretation:** These comparative statics map the theory to real markets. Bubbles are most likely when limits to arbitrage are tight, awareness is dispersed, and prices continue rising rapidly.

## 8 Detailed result map

- **Bubble setup.** Price exceeds post-crash value but continues growing.
- **Asymmetric awareness.** Arbitrageurs learn about the bubble at different times.
- **Higher-order uncertainty.** Each arbitrageur is unsure how many others know.
- **Coordination problem.** A critical mass must sell for the bubble to crash.
- **Riding equilibrium.** Rational arbitrageurs may remain invested despite recognizing mispricing.
- **Endogenous crash.** The crash occurs when enough informed arbitrageurs sell at the equilibrium sellout time.
- **Limits of efficiency.** Knowledge of mispricing does not imply immediate correction when arbitrage requires synchronized action.

## 9 Short summary

- Abreu and Brunnermeier explain how bubbles can survive in a market with rational arbitrageurs.
- The central friction is sequential awareness: arbitrageurs learn about the bubble at different times.
- Each arbitrageur knows the bubble exists but is uncertain about how many other arbitrageurs know.
- Since a critical mass is needed to burst the bubble, early arbitrage can lose money.
- Rational arbitrageurs may ride the bubble and exit only when enough others are likely to be aware.
- The model generates an endogenous crash time.
- The paper's contribution is to identify synchronization risk as a separate limit to arbitrage.

## 10 Conclusion

### 10.1 Core conclusion

Abreu and Brunnermeier show that asset-price bubbles can persist even when sophisticated arbitrageurs recognize overpricing. The reason is not that arbitrageurs are irrational, but that arbitrage requires synchronized action. When awareness of the bubble diffuses gradually, each arbitrageur is uncertain about whether enough others are ready to sell.

## 10.2 Economic intuition

The arbitrageur faces a timing problem. Selling too early can be costly because the bubble price may continue rising. Waiting too long is also risky because the crash can occur before she exits. The equilibrium balances these forces. Early in the bubble, riding is attractive; later, as awareness spreads, the risk of a coordinated selloff rises and exit becomes optimal.

## 10.3 Incremental contribution revisited

The paper's key incremental contribution is to introduce synchronization risk as a formal limit to arbitrage. This mechanism is distinct from noise-trader risk, fundamental risk, or short-sale constraints alone. It explains both the persistence of bubbles and the abruptness of crashes in one dynamic framework.

## 10.4 Implications for financial markets

The model helps explain why sophisticated investors may knowingly hold overpriced assets in technology booms, real estate booms, credit bubbles, or other asset-market manias. It also explains why a crash can happen suddenly after a long period of apparent stability: once the coordination threshold is reached, many arbitrageurs want to exit at the same time.

## 10.5 Limitations

The theory is stylized. It assumes a specified bubble price path, simplified position limits, and no explicit communication among arbitrageurs. It also does not directly estimate awareness diffusion. The value of the paper is therefore the mechanism rather than a complete empirical description of any specific bubble.

## 10.6 Final takeaway

The central takeaway is that market correction requires common timing, not just private information. Rational arbitrageurs can knowingly ride a bubble when they do not know whether enough others are ready to attack it.